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1. Introduction

1.1. The energy transition in Great Britain (GB) is marked by the rapid integration of intermittent renewable sources such as wind and solar power, which shapes the operational dynamics of the GB electricity market. The variability of these resources results in the complex task of maintaining the balance between supply and demand, a task carried out by the National Energy System Operator (NESO).

1.2. This project focuses on the prediction of the System Price, also known as the 'imbalance price', which reflects the real-time cost of balancing the difference between the contracted and actual generation or consumption of electricity in each half-hourly Settlement Period (SP). The System Price is paid by market participants (generators, suppliers) who have a net deficit from their contracted generation/consumption position, or paid to those with a net surplus from their contracted position.

1.3. The GB electricity market is rarely balanced through contracted positions alone, especially with the penetration of renewable energy where supply is highly variable and dependent on weather conditions. Consequently, the System Price often deviates from the Day-Ahead market price (wholesale price), creating both risk and opportunity (Figure 1).

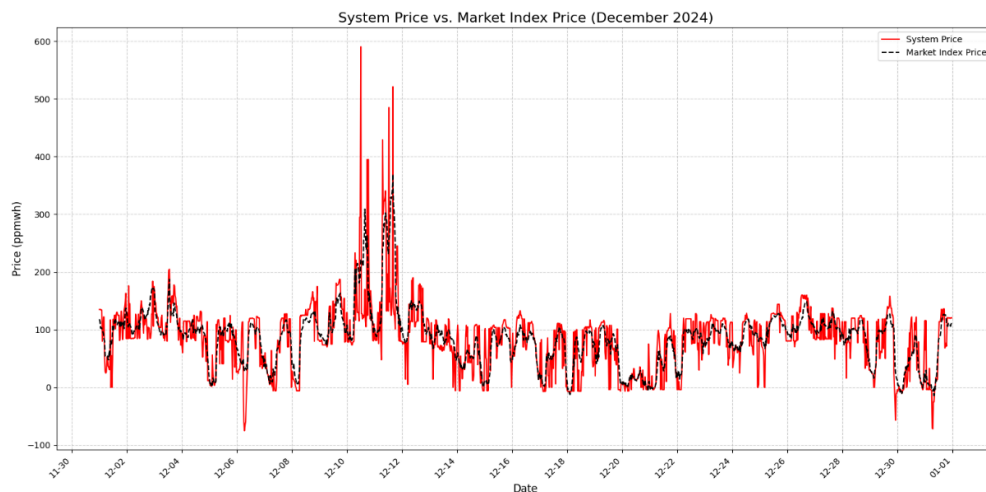


Figure 1. Comparison of System Price and Market Index Price in Dec 2024

1.4. The objective of this project is to evaluate the performances of three different models in the prediction of System Price, namely Multiple Linear Regression, Random Forest and XGBoost, and investigate which variables are impactful predictors on system price.

2. Literature Review

2.1. Overview of the GB Balancing Market

2.1.1. The GB Balancing Mechanism operates on 30-minute SPs, over which committed volumes are delivered and imbalance volumes are balanced. Participants must notify NESO of the volume they expect to generate or consume ('physical notification') one hour before each SP, a point known as 'gate closure'. By 'gate closure', flexible generators should also inform NESO of their offers/bids to increase/reduce generation/demand. NESO then accepts the most economic bids/offers to balance the system. The System Price reflects the costs that NESO incurs in balancing, since participants could deviate from their committed positions made at 'gate closure', due to a number of reasons such as demand forecasting error and variability in renewable production. If the Net Imbalance Volume (NIV) is positive, the system is short and NESO accepts more offers than bids (i.e. demand more than supply). If the NIV is negative, the system is long and NESO accepts more bids than offers (i.e. supply more than demand). Offers will be higher than the marginal generation costs as generators will increase output to sell power to the grid, while bids will be lower than the marginal generation costs as generators will reduce its own output and buy power from the system. Thus, the System Price will be higher or lower than marginal costs depending on whether the system is long or short in the given SP (Bunn, Inekwe and Macgeehan, 2021).

2.1.2. There is an incentive for participants to forecast the imbalance and intentionally aim to be out of balance in the opposite direction and profit from the difference between the System Price and the marginal cost of production. This practice is known as 'NIV-chasing' and is allowed under the Balancing and Settlement Code (Elexon, 2019). Changes in GB market rules from April 2019 onwards allowed participants to adjust their trading positions throughout the hour prior to the SP, thus speculation of System Price can be close to real time and participants can also incorporate data that only becomes available closer to the SP, although they cannot change their committed physical positions (Bunn, Inekwe and Macgeehan, 2021).

2.2. Existing research on forecast of System Price

2.2.1. There is a wealth of literature on wholesale price forecasting and analysis on the increased penetration of renewable energies (Weron, 2014; Salem *et al.*, 2019; Osoné and Kodaira, 2025), but comparatively sparse literature on imbalance price forecasting. We reviewed three key papers specific to the GB market. A broader horizon scan of literature utilising a range of models is included in Appendix A.

2.2.2. Bunn *et al.* (2021) examined system price forecasting across two timescales in the GB market: Day-Ahead and Intraday (up to 5 hours ahead), using Kernel Density Estimation and linear regression for the former, and a proprietary probabilistic forecasting tool for the latter. The linear regression method outperformed rudimentary proxies (day-ahead auction price and climatological prices) by 3%. The intra-day model performance had a Mean Absolute Error (MAE) of £21.1/MWh at 5 hours lead time, which was 5-40% more effective than the proxies.

2.2.3. Lucas *et al.* (2020) leveraged on the Loss of Load Probability (LOLP) data made available in UK for the first time in 2019 to forecast System Price. Three models were used: Gradient Boosting, Random Forest and XGBoost. XGBoost had a MAE of £7.89/MWh, R^2 of 76.8% and MSE of 124.74. The Random Forest model had a MAE of 6.97, R^2 of 80.4 and MSE of 105.38. As the NIV cannot be forecasted ahead of time with sufficient accuracy, the NIV variable engineered using quantile decomposition of historical NIV from settlement periods with similar market conditions. XGBoost was chosen over Random Forest as the best-performing model as it did not overwhelmingly rely on the estimated NIV as the dominant variable. The most influential variables in XGBoost were estimated NIV (28.6%), aggregated LOLP (27.5%), the month (14%) and aggregated De-rated margins (8.9%).

2.2.4. Bunn *et al.* (2021) found that a non-linear Markov Regime-Switching model outperformed a linear model and a Generalised Autoregressive Conditional Heteroskedasticity (GARCH) linear model in predicting the sign of the System Price in the GB market. The out-of-sample RMSE for the Regime-Switching method was 14.7 while the linear method was 15.4, while the in-sample R^2 of all 3 models are 38-39%. Notably, the data on prices did not undergo logarithmic transformation since the statistical price properties of interest could be masked by the transformation.

2.2.5. Statistical methods and time series models such as autoregressive integrated moving average (ARIMA) were often used to forecast short-term electricity prices in earlier studies (Weron, 2014; Rounkvist, Enevoldsen and Xydis, 2020). While time series models capture seasonality and linear patterns, they perform poorly with the non-linear and irregular trends inherent in System Prices, characterised by spikes and noise (Salem *et al.*, 2019; Yılan and Beykent, 2025). The non-constant mean and variance also complicate the process of forecasting (Lucas *et al.*, 2020). Majority of the literature agree that statistical models such as linear regression performs poorly due to the presence of extreme observations, and point to use of non-parametric techniques such as ensemble models (e.g. Random Forest and XGBoost) that are more flexible and adaptable to the complex dynamics and constraints exhibited by electricity price markets (Lucas *et al.*, 2020).

2.2.6. Most studies combine the use of linear regression models as a benchmark to compare with more sophisticated methods (Bunn, Inekwe and Macgeehan, 2021; Browell and Gilbert, 2022). Linear regression and autoregression models assume homoskedasticity, but electricity prices exhibit the characteristic of the variability of the series having a strong dependence on its own past, and often exhibits heteroskedasticity instead (Weron, 2014). Given the wide range of methodologies with varied performance, the selection of forecasting model must be informed by the data and market characteristics (Yılan and Beykent, 2025). A comparison between linear and non-linear models would thus be informative to understanding the GB market system prices.

3. Methodology

3.1. Data Selection

3.1.1. Data selections encompassed demand forecast attributes, generation attributes and exogenous factors such as temperature and market index prices. Datasets were sourced from various credited institutions such as Exelon, NESO, and the National Gas Transmission data portal. Details on raw datasets can be found in Appendix B. Depending on availability, data was retrieved via API or direct export and covered a four-year period from April 1, 2021, to April 1, 2025. The selected data period captures the lasting impact of the Russia-Ukraine war on energy prices globally (Bolton, 2025) and avoids severe deviations in the energy consumption patterns arising from COVID-19 lockdown, where demand reduced by up to 20% in the morning peak period in the UK (Anderson and James, 2021).

3.2. Data Processing and Cleaning

3.2.1. System price data is updated for every settlement period (Competition and Markets Authority, 2016). Based on this, all raw data sources were aligned by date and settlement period, with timestamps standardised using the format “yyyy-mm-dd-sp” to ensure consistency. Missing values were treated according to the statistical properties of each variable. Gas price and temperature observations were backfilled, as these variables exhibit relatively low volatility over half-hour settlement intervals. Missing data for market index prices or demand forecasts were excluded (including the associated merged variables), as imputing values could introduce bias and impact price dynamics such as price spikes, which is a major phenomenon our modelling aims to capture.

3.2.2. The resulting merged dataset was subsequently used for exploratory analysis and feature extraction. A time-based train–test split was conducted. The training set comprised of three years of data (April 1, 2021–April 1, 2024), while the test set consisted of one year of data (April 1, 2024–April 1, 2025). This split ensures that both datasets effectively capture seasonal patterns of the whole year.

3.3. Linear Regression Model

3.3.1. The first model, linear regression is a foundational statistical method, often considered the “original” regression model. Since Linear regression models are highly interpretable, it has been chosen as the benchmark model for this analysis. The formula is provided below:

$$Y=\beta_0+\beta_1X_1+\beta_2X_2+...+\epsilon$$

3.3.2. Classical Linear regression models rely on strict assumptions - linearity in parameters, exogeneity, random sampling of data, no multicollinearity, and homoscedasticity of error – which limit the ability of OLS models to accurately capture more complex and non-linear characteristics (Misiurek, Olkusi and Zyśk, 2025). To address the complexity of the features while still capturing critical outlier information, this study applied a two-pronged strategy, where the linear regression model was run using both transformed and untransformed data.

3.3.3. To evaluate model validity, Durbin-Watson test for autocorrelation, the Jarque-Bera test for normality, the Breusch-Pagan test for heteroscedasticity, and the Rainbow test for linearity were applied to determine if linear regressions assumptions were satisfied and glean if a linear regression model was an appropriate model for forecasting system prices. Lasso regression was explored to mitigate risk of overfitting since it typically produces estimates in which some of the coefficients are set exactly to zero, thereby performing a feature selection (Ludwig, Feuerriegel and Neumann, 2015).

3.4. Random Forest Model

3.4.1. Random Forest (RF) was selected as a non-linear benchmark to complement the linear regression models, due to its ability to capture complex interactions and non-linear relationships without requiring explicit feature engineering (Ludwig et al., 2015). Unlike linear models, RF constructs an ensemble of decision trees that approximate non-linear thresholds directly from the data, making it well-suited to modelling electricity price dynamics.

3.4.2. To optimise the balance between predictive accuracy and generalisation, model development utilised grid search optimisation technique that exhaustively evaluates a manually specified subset of the hyperparameter space as shown in Table 1. To reduce correlation between individual trees, the number of features considered at each split was fixed to the square root of the total feature set. The selection process prioritised configurations that maximised the test R^2 score while maintaining a minimum sample threshold per leaf to ensure every terminal node remained statistically significant.

Table 1 Hyperparameter Search Space for RF and XGBoost

Model	Parameter	Values Tested	Purpose
RF	Estimators	200, 400	Ensemble Size
	Max Depth	12, 18, None	Complexity
	Min Samples Leaf	10, 25, 50	Regularisation
XGBoost	Learning Rate	0.01, 0.05, 0.1	Step Size
	Max Depth	4, 6, 9	Structural Depth

	Gamma	0, 1, 5	Pruning Threshold
	Reg Alpha (L1)	0, 10, 50	Sparsity
	Min Child Weight	10, 20	Leaf Robustness

3.5. XGBoost Model

3.5.1. The final regression model, XGBoost, was selected because it is another tree-based ensemble learning algorithm that has gained prominence in electricity price forecasting due to its capabilities in modelling nonlinear and complex temporal dependencies (Yılan and Beykent, 2025).

3.5.2. Consistent with the grid search methodology applied to the RF model, a systematic hyperparameter sweep was conducted to evaluate the search space in Table 1. However, unlike RF, the XGBoost search space was expanded to include regularisation parameters specific to gradient boosting, like L1 regularisation (reg_alpha) and Gamma pruning, alongside Learning Rate (0.01–0.3) and Max Depth (3–9).

3.5.3. While the RF optimisation focused primarily on maximising predictive power, the XGBoost selection strategy was refined to prioritise stability. Given the high variance observed in the training data, we explicitly monitored the divergence between Training and Test metrics, rejecting configurations where the overfitting gap exceeded 0.20 to prevent the model from memorising noise.

3.6. Feature Engineering - Domain Logic Filtering

3.6.1. Before applying statistical tests, the raw variable set was evaluated against domain knowledge to identify features that would compromise model validity and reduced the number of features to 16 (including the target variable) down from the original 34. Reduction is summarised in Appendix B.

3.6.2. A critical priority was the removal of variables that are determined simultaneously with the target variable (System Price), consistent with the practice in (Lucas et al., 2020), as including these would artificially inflate predictive performance without offering genuine forecasting utility. Consequently, NIV and accepted volume data were excluded.

3.6.3. Several variables represented identical underlying phenomena. To prevent immediate multicollinearity, we retained Demand Forecast as the primary load indicator and removed its highly correlated subsets, specifically Transmission System Demand, England & Wales Demand, and Net Demand (Figure 2). Similarly, Embedded Solar Capacity and Embedded Wind Capacity were dropped as they are static or low variance figures that do not capture the dynamic, time-dependent nature of price formation compared to actual generation forecasts.

3.6.4. Additionally, to reduce dimensionality while preserving signal, Wind Offshore and Wind Onshore were merged into a consolidated wind generation feature. The resultant correlation heatmap showed that collinearity was reduced (Figure 3).

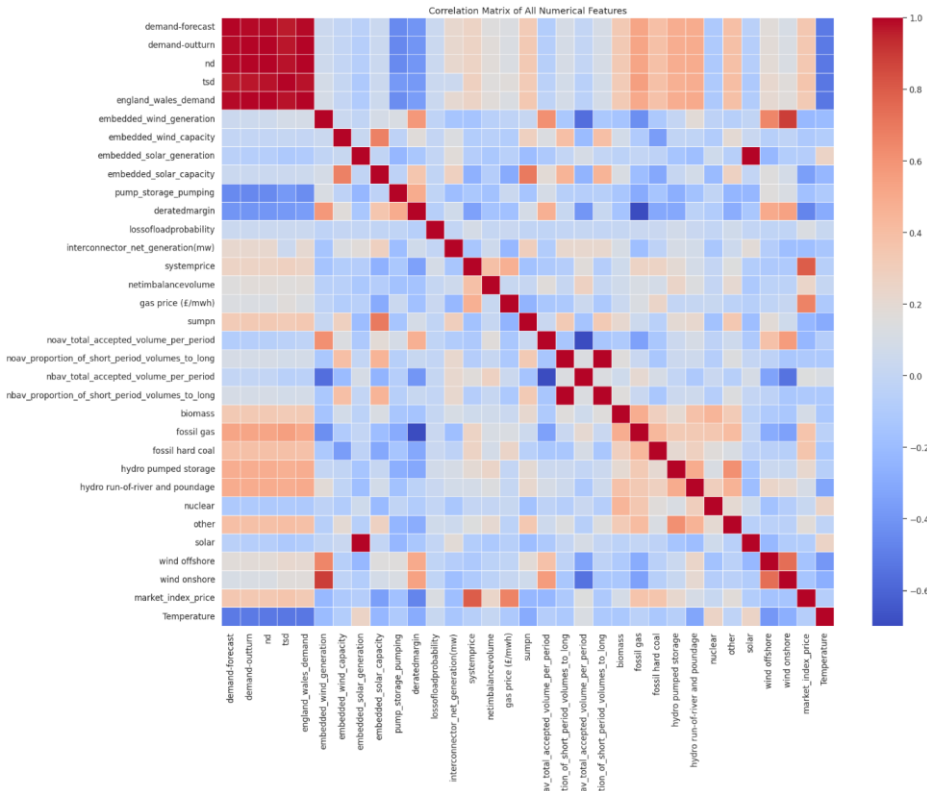


Figure 2. Correlation Heatmap of all features before dropping correlated and/or redundant features

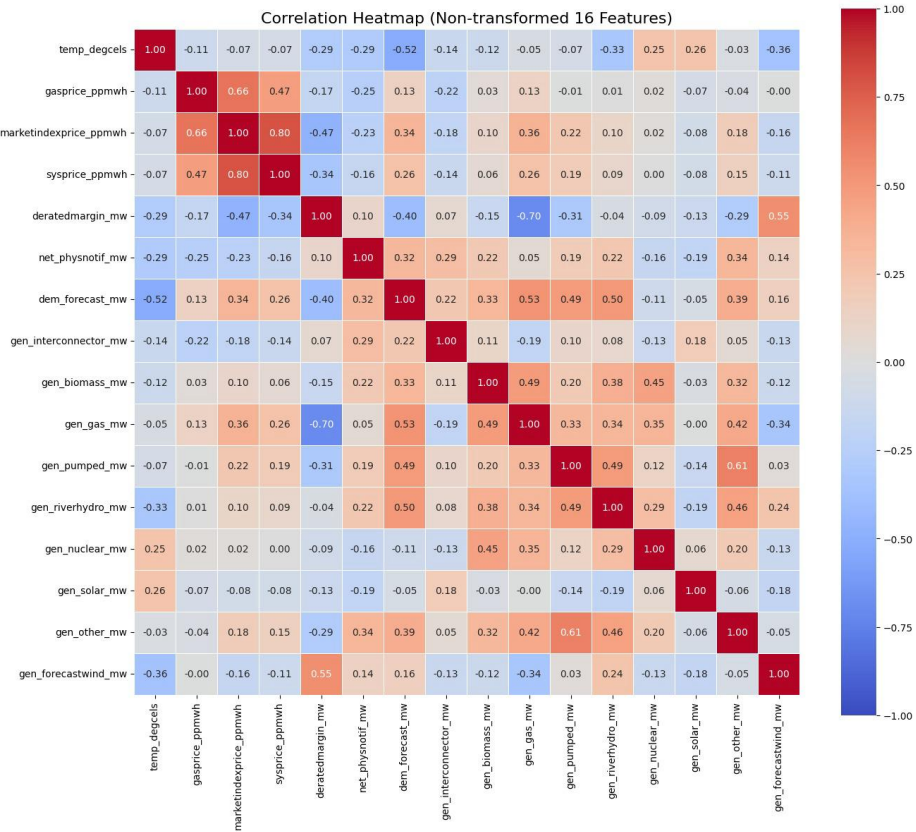


Figure 3. Correlation Heatmap of features after reduction of features

3.7. Feature Engineering – Transformations and Ordinary Least Squares (OLS) Model

3.7.1. Following the initial exclusion of variables based on domain logic, the remaining feature set underwent a detailed distributional analysis to address non-normality and scale disparities. Feature transformations were explored to mitigate violations of strict OLS assumptions, such as normality and linearity, with the aim of establishing a more precise benchmark OLS model. Figure 4 displays the behaviour of System Price data (the target variable). This variable exhibit extreme positive skewness driven by a long right tail, reflecting that system stress conditions lead to very high prices. While standard statistical practice typically prescribes logarithmic transformation to normalise such residuals, no transformations were applied and the variable was retained in its original units (£/MWh). Consequently, the analysis accepts the violation of the normality assumption in the OLS model to preserve the economic interpretability and ensure that scarcity events remain appropriately represented in the model.

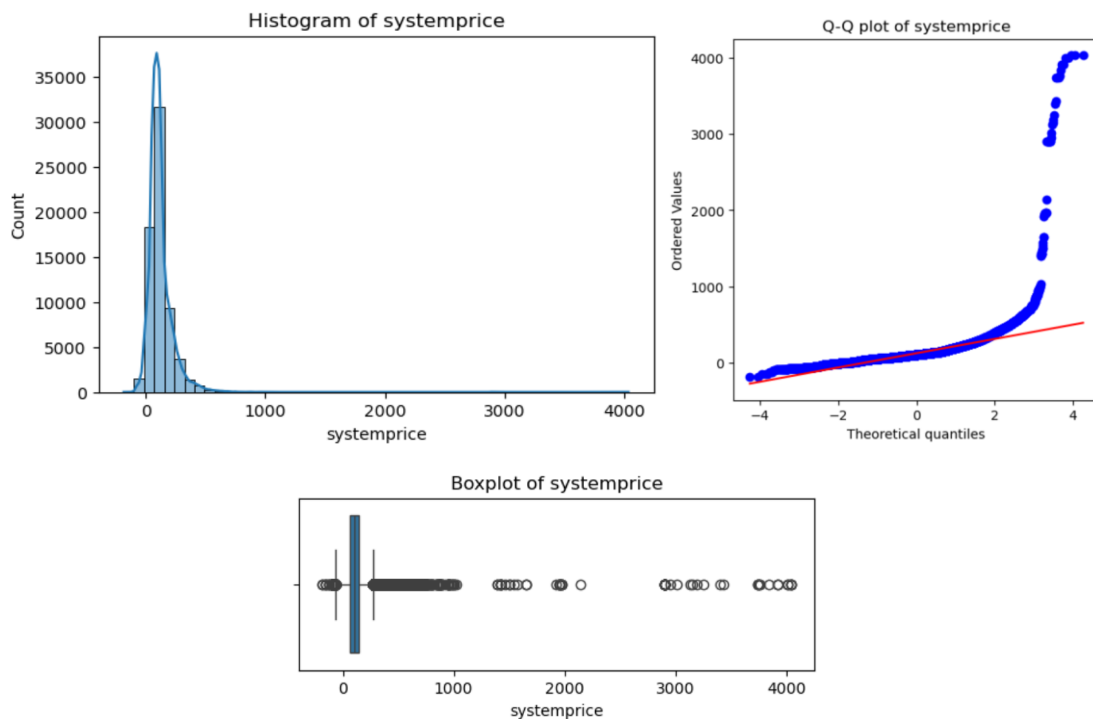


Figure 4. Histogram, Box plot and Q-Q plot of System Price (Target Variable)

3.7.2. For the predictor variables, a tailored transformation strategy was employed to handle the "zero-inflated" nature of certain generation types. Variable such as gas price, biomass, fossil gas, and pumped storage displayed distributions characterised by a massive concentration of values at zero, representing periods of inactivity, followed by a continuous distribution of active generation (Figure 5). To stabilise the variance of the magnitude component and mitigate the influence of extreme outliers, we applied a logarithmic transformation to these datasets. A constant of 1 was added to ensure zero values were captured, defined formally as:

$$xlog' = \ln x + 1$$

Where x represents the raw generation value in MW or price in £/MWh.

3.7.3. For variables exhibiting milder skewness ($0.5 < \text{Skewness} < 1.0$), such as Total Wind Forecast, a less aggressive square root transformation was applied to approximate normality without over-compression (Figure 6):

$$xsqrt' = x$$

3.7.4. Figure 7 illustrates the correlation matrix with 16 variables with the transformations incorporated.

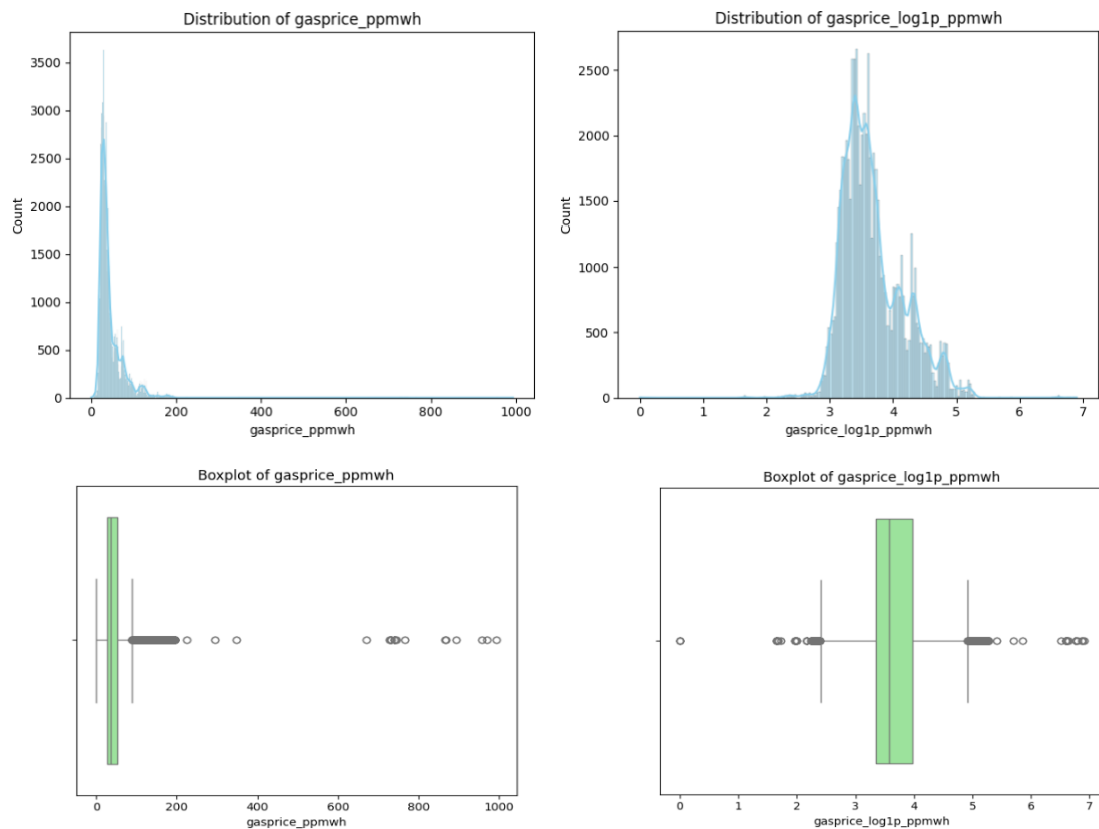


Figure 5. Log1p transformation of Gas Price - Pre-transformation (Left), Post-transformation (Right)

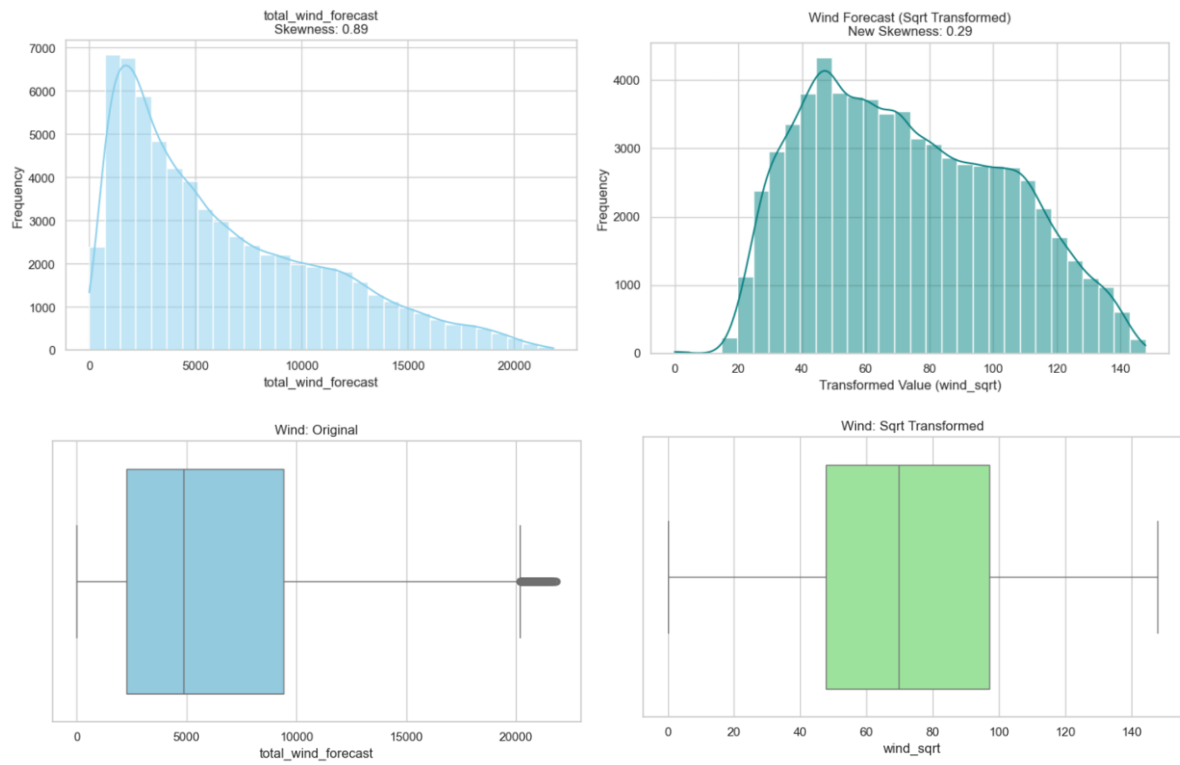


Figure 6. Square root transformation of total wind forecast - Pre-transformation (Left), Post-transformation (Right)

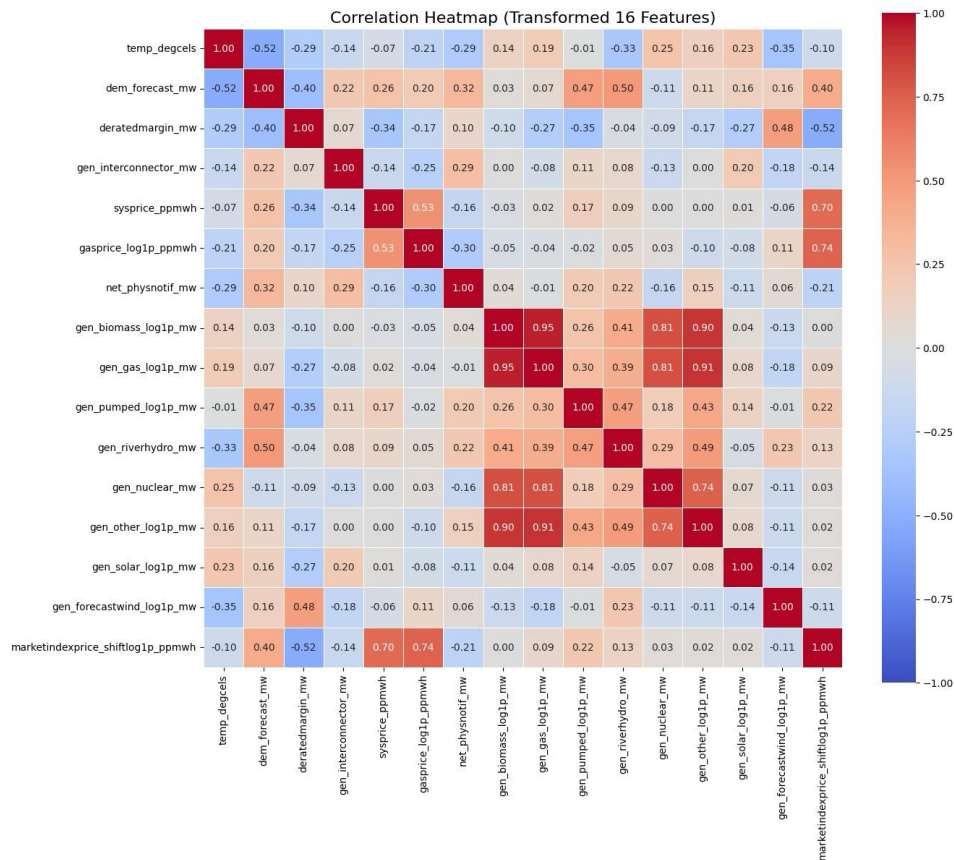


Figure 7. Correlation Heatmap of variables post-transformation (target variable: sysprice_ppmwh)

3.8. Feature Scaling

3.8.1. To ensure uniform contribution to the model optimisation processes, all continuous predictors were standardised using Z-score scaling after splitting the train-test data. This transforms the features to a common scale with a mean (μ) of 0 and standard deviation (σ) of 1:

$$z = \frac{x - \mu}{\sigma}$$

4. Results

4.1. Linear Regression Results

4.1.1. The linear model with transformed data achieved a modest performance in terms of test R^2 and MAE (Table 2). The higher train R^2 and MAE indicates generalisation issues, since the model explains about 50.4% of variance in training, but only 33.1% in testing. A closer examination of Figure 8 and Figure 9 shows that the model generalised well in the normal ranges but overpredicted dips and underpredicted spikes in System Prices.

4.1.2. Despite efforts to address multicollinearity and non-normality at data preparation stage, the linear model still exhibited violations of OLS assumptions of positive autocorrelation (Durbin-Watson = 0.878 versus the ideal value of 2), extreme non-normality of residuals (Jarque-Bera p-value = 0.00, high skew and kurtosis), heteroscedasticity (Breusch-Pagan p-value = 0.000) and non-linearity (Rainbow Test p-value = 0.00) – refer to Figure 10 for residual diagnostics. LASSO regularisation led to a marginal improvement in test R^2 and MAE (Appendix C).

Table 2. Performance metrics for OLS (Transformed) model

	Train			Test		
R²	0.5040			0.3308		
MAE	43.12			26.73		
	coef	std err	t	P> t	[0.025	0.975]
const	134.2622	0.408	328.954	0.000	133.462	135.062
temp_degcel	0.4015	0.721	0.557	0.578	-1.012	1.815
dem_forecast_mw	-0.7893	0.908	-0.869	0.385	-2.569	0.991
deratedmargin_mw	3.7217	0.909	4.094	0.000	1.940	5.504
gen_interconnector_mw	-6.3795	0.507	-12.587	0.000	-7.373	-5.386
gasprice_log1p_ppmwh	0.0749	0.757	0.099	0.921	-1.408	1.558
net_physnotif_mw	-1.7323	0.497	-3.486	0.000	-2.706	-0.758
gen_biomass_log1p_mw	4.6838	1.696	2.762	0.006	1.360	8.008
gen_gas_log1p_mw	-31.6018	2.097	-15.070	0.000	-35.712	-27.492
gen_pumped_log1p_mw	4.4408	0.569	7.804	0.000	3.325	5.556
gen_riverhydro_mw	-0.2864	0.624	-0.459	0.646	-1.509	0.936
gen_nuclear_mw	0.3299	0.796	0.414	0.679	-1.230	1.890
gen_other_log1p_mw	21.1848	1.332	15.908	0.000	18.575	23.795
gen_solar_log1p_mw	0.4271	0.471	0.907	0.365	-0.496	1.350
gen_forecastwind_log1p_mw	-3.7742	0.598	-6.315	0.000	-4.946	-2.603
marketindexprice_shiftlog1p_ppmwh	91.1308	0.811	112.380	0.000	89.541	92.720
=====						
Omnibus:	103893.183	Durbin-Watson:		0.878		
Prob(Omnibus):	0.000	Jarque-Bera (JB):		707194234.640		
Skew:	17.749	Prob(JB):		0.00		
Kurtosis:	585.679	Cond. No.		13.0		

Seasonal Performance in Apr 2024 - Apr 2025: Actual vs. Predicted GB Electricity Prices (OLS Transformed data)

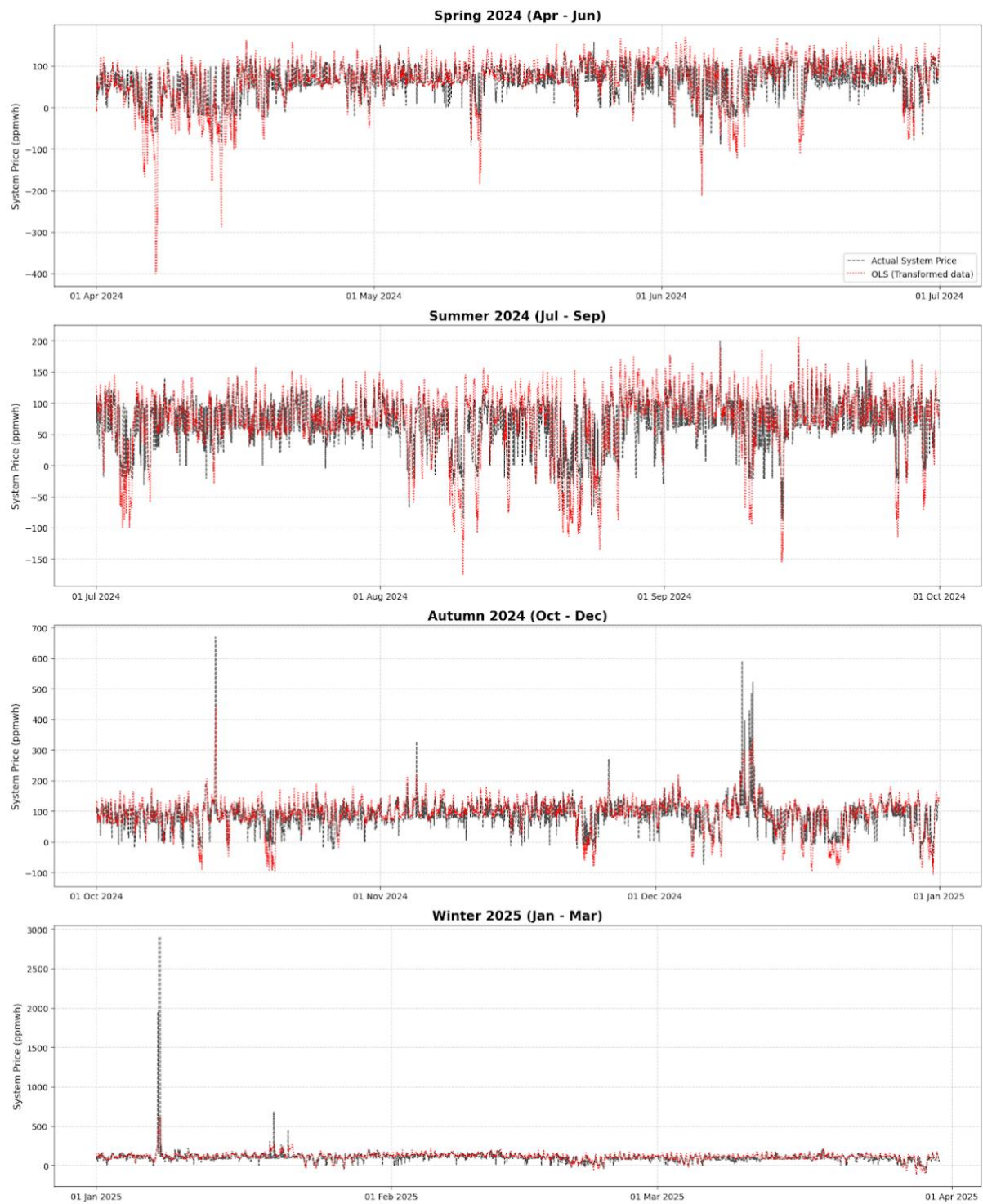


Figure 8. Seasonal plot of actual vs predicted System Price for OLS (Transformed Model)

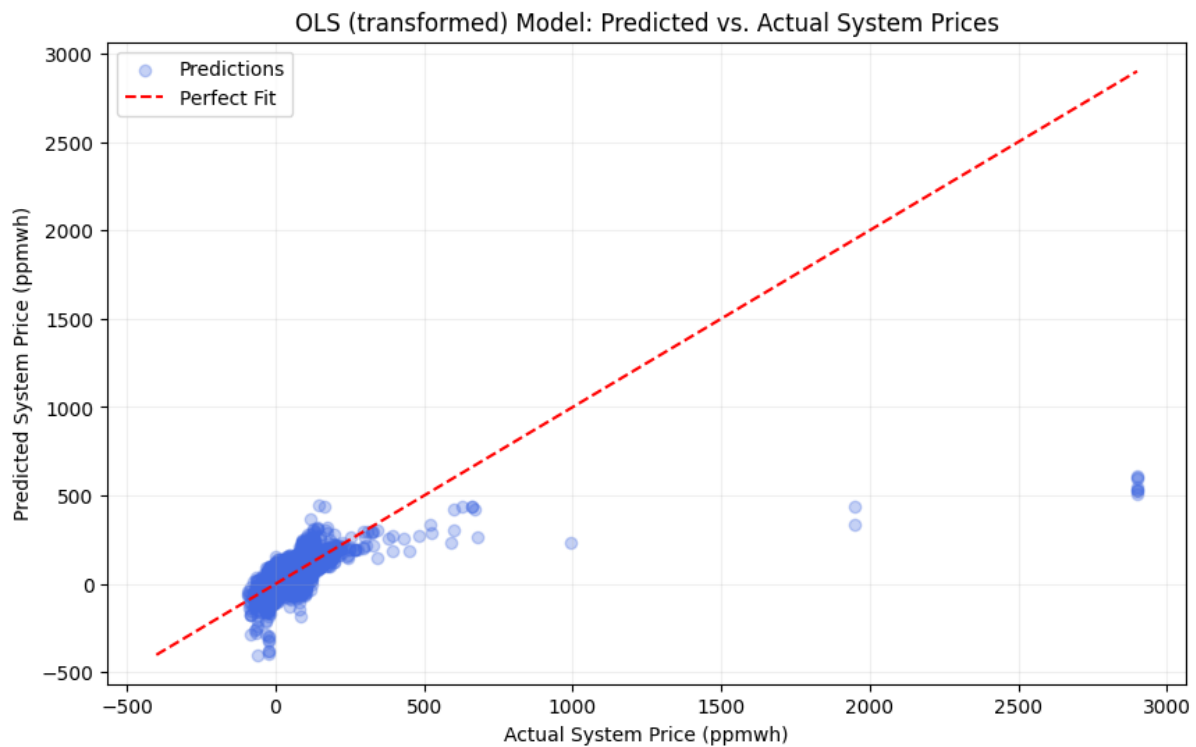
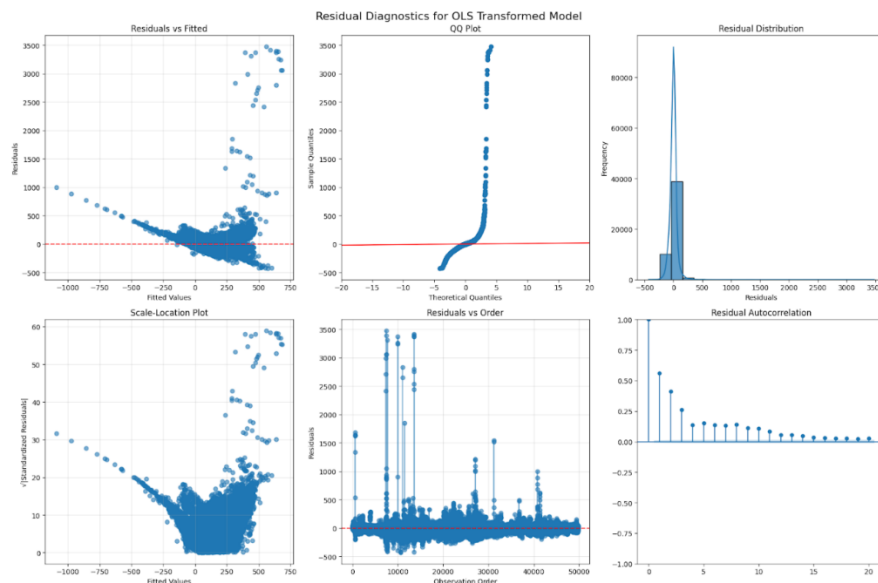


Figure 9. OLS (Transformed): Predicted vs. Actual System Prices



Durbin-Watson: 0.878

Interpretation: Values near 2 indicate no autocorrelation

< 1.5: Positive autocorrelation

1.5–2.5: No autocorrelation

> 2.5: Negative autocorrelation

Jarque-Bera Test: Stat = 707194234.640, p-value = 0.000

Skewness: 17.749, Kurtosis: 585.679

H0: Residuals are normally distributed

Conclusion: Reject H0 (non-normal)

Breusch-Pagan Test: LM = 1058.906, p-value = 0.000

H0: Homoscedasticity (constant variance)

Conclusion: Reject H0 (heteroscedastic)

Rainbow Test: F = 1.045, p-value = 0.000

H0: Model is correctly specified (linear)

Conclusion: Reject H0 (non-linear)

Figure 10. Residual diagnostics of OLS (Transformed) model

4.2. Random Forest Results

4.2.1. A targeted hyperparameter sweep was performed for the RF model using the non-transformed dataset, varying the number of estimators, maximum tree depth, and minimum samples per leaf (details in Appendix D). Model selection was guided by test MAE, stress-event MAE, and the training–test R^2 gap to balance predictive accuracy and generalisation.

4.2.2. The sweep showed that deeper trees reduced training error but increased overfitting, while overly shallow configurations led to higher test error (Table 3). The final configuration, using 400 estimators, a maximum depth of 18, and a minimum leaf size of 10, achieved the most stable out-of-sample performance with improved robustness during stress events and was therefore selected for final evaluation.

Table 3. Selected Results of RF using various hyperparameter configurations

	n_estimators	max_depth	min_samples_leaf	Train_MAE	Test_MAE	Train_R2	Test_R2	O0B_R2	Stress_MAE	Not_Stress_MAE	Overfit_Gap
7	200	None	25	35.728744	22.703230	0.644499	0.475326	0.596277	84.257255	21.093005	0.169172
16	400	None	25	35.657272	22.629828	0.644152	0.475721	0.596177	84.137917	21.020804	0.168431
13	400	18	25	35.771853	22.594187	0.644105	0.478350	0.595247	84.852454	20.965539	0.165754
10	400	12	25	37.521594	22.854404	0.637527	0.479626	0.595650	84.114296	21.251873	0.157901
0	200	12	10	35.499003	22.683936	0.706194	0.509650	0.631635	82.097168	21.129713	0.196543
9	400	12	10	35.495146	22.664133	0.708018	0.524782	0.632537	80.851168	21.141986	0.183236
6	200	None	10	30.301384	22.306974	0.731418	0.528072	0.637699	82.583246	20.730175	0.203346
3	200	18	10	30.980375	22.234338	0.732118	0.538979	0.639679	82.850863	20.648638	0.193139
12	400	18	10	30.986049	22.294090	0.730559	0.541101	0.638424	81.568790	20.743491	0.189457
15	400	None	10	30.269212	22.270521	0.732248	0.547550	0.637441	80.306515	20.752326	0.184697

4.2.3. The tuned RF model showed improved generalisation relative to the baseline configuration (Table 4). While training MAE increased following regularisation, test MAE decreased and the training–test R^2 gap narrowed, indicating reduced overfitting. The predicted versus actual price plot (Figure 11) demonstrates good alignment for typical price ranges as predictions cluster near the identity line for typical market conditions, with deviations increasing at higher prices.

Table 4. Random Forest Performance Summary

Model	Train MAE (£/MWh)	Test MAE (£/MWh)	Train R^2	Test R^2
Base Random Forest (using default hyperparameters)	12.68	22.14	0.9551	0.5805
Random Forest (using tuned hyperparameters)	30.99	22.29	0.7306	0.5411

4.2.4. However, as with XGBoost, the RF model systematically underestimates the magnitude of extreme price spikes. While high-price events are detected, their severity is dampened in the predictions, reflecting the model's tendency to average across ensemble trees and favour central tendencies in the data.

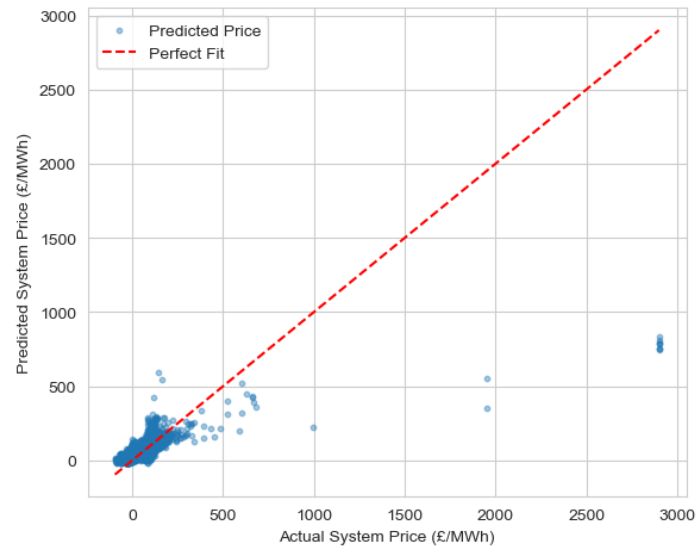


Figure 11. Predicted Price vs Actual Price for Random Forest

4.3. XGBoost Results

4.3.1. A targeted hyperparameter sweep was conducted for the XGBoost regressor using the non-transformed dataset, varying learning rate, tree depth, gamma, L1 regularisation strength, and minimum child weight (details in Appendix E). Model selection prioritised test MAE, stress-event MAE, and the training–test R^2 gap to balance predictive accuracy and generalisation.

4.3.2. The sweep revealed a clear trade-off between model complexity and overfitting (Table 5). Deeper trees and weaker regularisation achieved higher training R^2 but exhibited reduced generalisation, while more constrained configurations increased bias. The final model, with maximum depth 6, learning rate 0.01, gamma 5, L1 regularisation of 10, and minimum child weight of 10, achieved the lowest test MAE while maintaining stable out-of-sample performance and was retained as the final XGBoost model.

4.3.3. The optimised XGBoost model achieved lower test MAE and higher test R^2 compared to the baseline model (Table 6). Although training R^2 decreased after regularisation, test performance improved, suggesting enhanced out-of-sample generalisation. Predicted versus actual prices (Figure 12) closely follow the identity line under normal market conditions, with increasing dispersion at higher prices.

4.3.4. As with the RF model, XGBoost underestimates the magnitude of extreme price spikes. While high-price events are generally identified, their predicted values remain compressed relative to observed prices, indicating reduced accuracy during periods of extreme market stress.

Table 5. Selected Results of XGBoost using various hyperparameter configurations

	n_estimators	max_depth	learning_rate	min_child_weight	gamma	reg_alpha	Train_MAE	Test_MAE	Train_R2	Test_R2	Stress_MAE	not_stress_MAE	Overfit_Gap
51	1000	9	0.01	10	5	10	28.080376	21.303429	0.897726	0.682290	73.246086	19.944633	0.215436
49	1000	9	0.01	10	1	10	28.085253	21.290807	0.897625	0.680784	73.209700	19.932632	0.216841
50	1000	9	0.01	10	5	0	28.143914	21.315107	0.897102	0.680757	73.626621	19.946662	0.216345
48	1000	9	0.01	10	1	0	28.126085	21.320334	0.897310	0.679792	74.214571	19.936645	0.217518
25	1000	6	0.01	10	1	10	33.999201	21.048723	0.844667	0.663137	69.529623	19.780485	0.181530
27	1000	6	0.01	10	5	10	33.999207	21.048730	0.844667	0.663137	69.529620	19.780492	0.181530
11	1000	4	0.05	10	5	10	33.094223	21.800543	0.864298	0.662890	74.284224	20.427594	0.201408
9	1000	4	0.05	10	1	10	33.094223	21.800543	0.864298	0.662890	74.284224	20.427594	0.201408
26	1000	6	0.01	10	5	0	33.984382	21.174209	0.844608	0.660187	69.230329	19.917083	0.184421
24	1000	6	0.01	10	1	0	33.984382	21.174209	0.844608	0.660187	69.230329	19.917083	0.184421

Table 6. XGBoost Performance Summary

Model	Train MAE (£/MWh)	Test MAE (£/MWh)	Train R ²	Test R ²
Base XGBoost (using default hyperparameters)	28.41	23.83	0.9095	0.5591
XGBoost (using tuned hyperparameters)	33.99	21.05	0.8447	0.6631

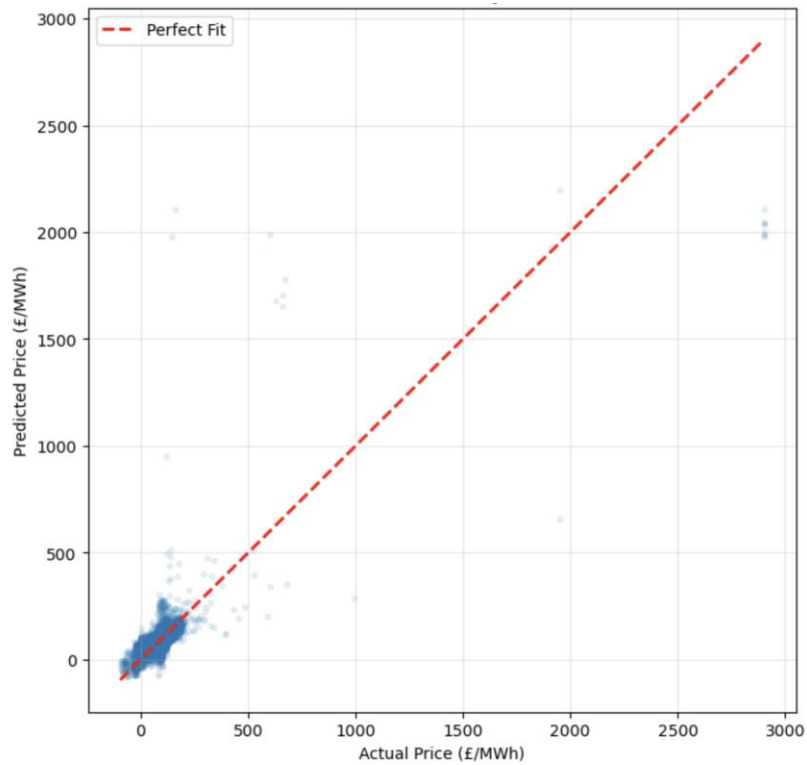


Figure 12. Predicted Price vs Actual Price for XGBoost

5. Discussion

5.1. General model performance

5.1.1. When considering overall model performance, XGBoost performed the best out of the three models, with the lowest MAE of £21.05/MWh and highest R^2 of 0.6631 (Table 7). This is followed by RF with a MAE of £22.29/MWh and an R^2 of 0.5411, then OLS with a MAE of £26.73/MWh and an R^2 of 0.3308. Typical wholesale electricity in stable periods trades at £40–£80/MWh, meaning XGBoost's error represents roughly 25–50% of normal price levels, whilst OLS's error reaches 33–65%.

5.1.2. Whilst R^2 is widely used as the standard model performance metric in academic literature, it can be unreliable for assessing the performance of non-linear models (Spiess and Neumeyer, 2010). Because of this, MAE is also used alongside R^2 as a secondary, more robust measure for inter-model comparison. Whilst R^2 and MAE produce the same ranking (XGBoost > RF > OLS), R^2 substantially exaggerates the performance gap. By R^2 , RF appears far inferior to XGBoost; by test MAE, the actual difference is modest (£1.24/MWh).

5.1.3. The ordering of these models' general performance is consistent with what would be expected given existing literature. XGBoost would perform the best as is it adaptable to complex dynamics in a stressed market and is not overly influenced by a single dominant variable (Lucas *et al.*, 2020). RF would perform better than OLS, as non-parametric models are known for being better than OLS at modelling complex non-linear systems (Bunn, Inekwe and Macgeehan, 2021; Browell and Gilbert, 2022).

Model	Train MAE (£/MWh)	Test MAE (£/MWh)	Train R^2	Test R^2
Optimised XGBoost (Untransformed)	33.99	21.05	0.8447	0.6631
Optimised RF (Untransformed)	30.99	22.29	0.7306	0.5411
OLS (Transformed)	43.12	26.73	0.5040	0.3308

Table 7. Comparison of models' performance, ordered from lowest to highest Test MAE

5.2. Analysis of performance of modelling System Price spikes

5.2.1. Electricity prices are nonlinear and characterised by spikes and noise (Lucas *et al.*, 2020). The correct prediction of system price spikes is particularly important for energy traders who engage in a strategy known as ‘NIV-chasing’, as they can generate large amounts of revenue during times of high system price volatility (Bunn, Inekwe and Macgeehan, 2021).

5.2.2. We defined spike as any system price over £150/MWh, which is approximately 1.5 times the interquartile range from the mean system price of the test data. XGBoost again emerged as the top performer in predicting spikes, with the lowest MAE (Stress) of £69.53 and highest F1 score of 0.5096 (Table 8, Figure 13). This finding is aligned with literature review since XGBoost’s boosting algorithm enables the sequential trees to correct the residuals from previous trees, thus capturing the complex, non-linear ‘hockey stick’ price curve (Lucas *et al.*, 2020).

Model	MAE (Normal £0–150)	MAE (Stress >£150)	Precision	Recall	F1
Optimised XGBoost (Untransformed)	19.78	69.53	0.5056	0.5137	0.5096
OLS (Transformed)	22.10	78.73	0.3180	0.7303	0.4431
Optimised RF (Untransformed)	20.74	81.57	0.5016	0.3618	0.4204

Table 8. Comparison of models' performance in predicting system price spikes, ordered from highest to lower F1 scores

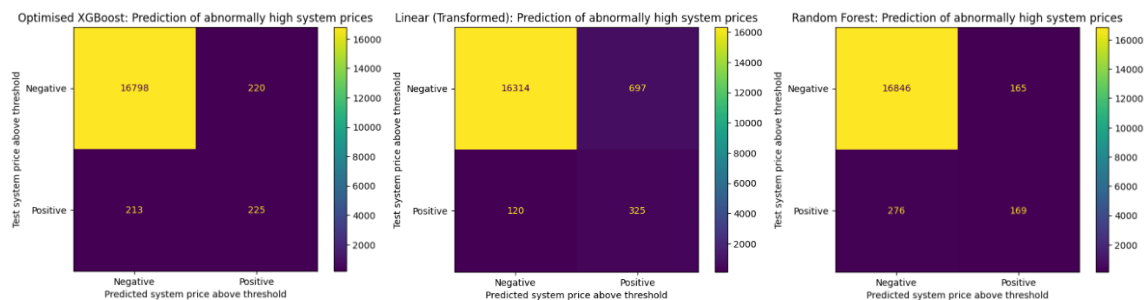


Figure 13. Confusion Matrix for all three models

5.2.3. In comparison, the RF and OLS models underpredicted at System Price spikes. One possible reason is that the minimisation of sum of squared errors caused the OLS model to prioritise predicting non-extreme prices occurring majority of the time and performed poorly during rare price spikes. The linear relationship failed to capture the exponential increase in price spikes (Figure 9). The RF model demonstrated a "ceiling effect" (Figure 11) because it relies on averaging the values of leaf nodes and is mathematically incapable of extrapolating beyond the range of values observed in the training leaves. The model dampens extreme volatility (the predictions rarely exceeded £300/MWh) during system stress events where actual prices spiked above £500/MWh, resulting in a high MAE (stress) of £81.57.

5.3. Feature importance

5.3.1. A comparative analysis of feature importance across the RF and XGBoost models reveals a fundamental divergence in how both algorithms model system prices. While both models agree on the primary cost drivers, however they diverge significantly in how they interpret secondary physical signals (Figure 14).

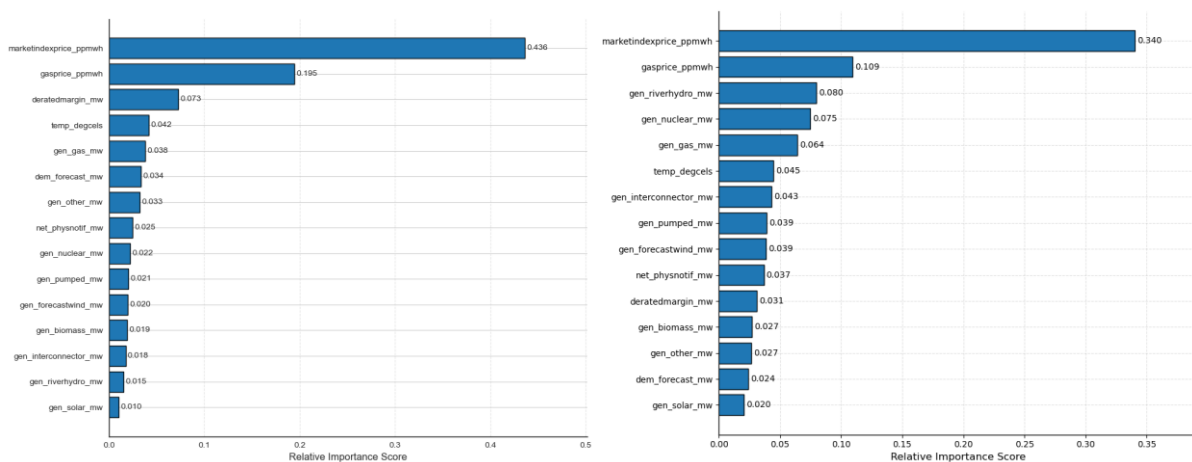


Figure 14. Feature Importance for Random Forest (Left), XGBoost (Right)

5.3.2. Consistent with the findings of Lucas et al. (2020), the Market Index Price (MIP) was the most dominant variable across all three models. This is to be expected, as MIP is used in the calculations for System Price (Elexon, 2025a). Inclusion of MIP likely reduces the importance of other non-price features, as it will already contain a significant amount of information relevant to System Price that would otherwise be gleaned from other individual features.

5.3.3. Gas prices also had a major influence in RF and XGBoost, as it is ranked as the second-best performing feature in both models. This is likely due to gas setting the wholesale price of electricity in the UK 98% of the time (Zakeri *et al.*, 2023) under the merit order system in the UK, where the most expensive form of generation running at a given time sets the market price.

5.3.4. A notable distinction arises in the other secondary features. The RF model prioritised De-rated Margin and Temperature, while the XGBoost model is in favour of direct generation resources (river hydro, nuclear and gas). The RF model's reliance on De-rated Margin and Temperature indicates that the model can identify conditions where the system is under stress and high System Prices may occur. De-rated Margin acts as a scarcity signal since a low margin indicates a high Loss of Load Probability (LOLP), which might trigger System Price spikes. Temperature also correlates with demand side stress due to increased energy demand (Hekkenberg *et al.*, 2009). However, because RF relies on proxies (de-rated margin and temperature) rather than the actual generation like XGBoost, it struggled to predict spikes to its true magnitude, contributing to the "ceiling effect" (Zakariah, 2014) .

5.3.5. In contrast, XGBoost's selection of Hydropower (River Hydro), Nuclear, and Gas Generation suggests it has learnt the operational mechanism of the supply stack. By focusing on the actual generation output, XGBoost could identify the unit setting the marginal price and actively model the merit order effect. This is a possible explanation for XGBoost's superior performance in predicting price spikes by achieving a lower overall stress MAE (£69.53) compared to the RF model (£81.57) - Table 8.

5.4. Limitations And Constraints

5.4.1. A notable concern is that that data distribution for our train and test sets differs quite significantly (Figure 15). To avoid data leakage, we trained the model using data from April 2021 to April 2024, during the post-Ukraine energy crisis. Over this period, System Prices were highly volatile and reached above £1,000/MWh several times (Elexon, 2021; Matson, 2023), whereas prices were more stable between 2024 and 2025. Because of this, it is likely that the generalisability of our models to the test data was reduced, as the models would have trained on patterns in the training data that are fundamentally different than what is present in the test data.

5.4.2. To mitigate the impact of this distributional shift without sacrificing sensitivity to price spikes, future iterations should prioritise separating volatility caused by “crisis” effects from volatility seen in a typical market. A regime-switching framework could be employed (Kapoor, Wichitaksorn and Zhang, 2023). This approach would allow the system to learn distinct predictive functions for 'nominal' and 'stress' states, thereby preserving the capability to forecast extreme values, while preventing the impacts of unrepresentative training data.

5.4.3. Another limitation of the current methodology is the presence of look-ahead bias, as the model relies on input data not necessarily accessible at the time of inference. To simplify the data preparation step of this study, not all variables accounted for true data availability. For instance, the model included observed temperature data recorded during the settlement period, rather than the forecasts available beforehand. This may have limited the model's ability to simulate real-time price prediction. Future studies should substitute these with forecasts available to traders at time of prediction or lagged data to better reflect reality.

5.4.4. It is also worth exploring if there are systematic differences in how well the model performs at different times of year (seasonally), or during periods of known high or low market stress. The value of this type of evaluation is heightened because of increasing levels of renewable generation as a consequence of decarbonising policy measures in the GB electricity market. It could then translate to having models tailored to each season with different parameters and variables.

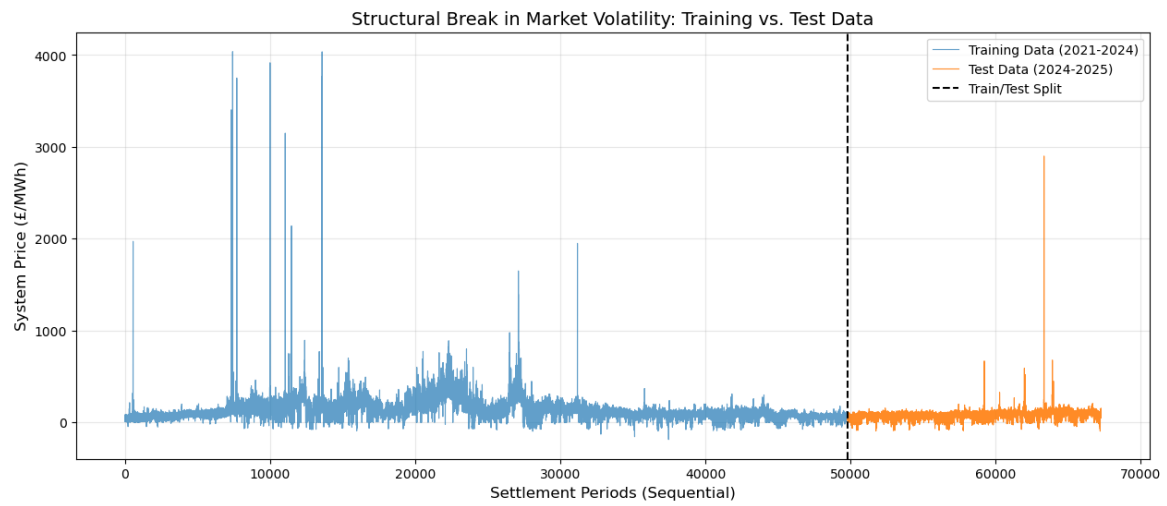


Figure 15. Actual System Price over train and test period

6. Conclusion

6.1. With the growing complexity of electricity markets and volatility of energy supply due to the increased penetration of renewables, the forecasting of System Price and imbalance is an essential tool for market participants and system operators alike. This paper sought to evaluate the performances of Linear Regression (OLS), Random Forest and XGBoost in the prediction of System Price. XGBoost was the top performer with the lowest MAE of £21.05/MWh and highest R^2 of 0.6631, a performance that is on par with other published papers (Bunn, Inekwe and Macgeehan, 2021). XGBoost also out-performed in the prediction of price spikes, with a MAE (stress) of £69.53/MWh and F1 score of 0.51. This corroborates with existing literature that non-parametric models are better than linear regression at modelling complex non-linear relationship of electricity prices.

6.2. In terms of investigating which variables are impactful predictors, this study found that Market Index Price was the most influential feature across all three models, while Gas Price was the second most influential in two out of three of the models. This key insight helps market participants prioritise which information they should consider in their analysis, especially since there is a burgeoning amount of data made available as the delivery settlement period approaches.

6.3. This study contributes to the growing volume of work on imbalance forecasting. Future research could shed light on improving prediction accuracy, which is of great importance to maximise revenue for NIV chasers and battery operators (Bunn, Inekwe and Macgeehan, 2021). For example, regime-switching models and concept drift detection tools could be deployed to identify when markets switch modes from normal to stress conditions (Kapoor, Wichitaksorn and Zhang, 2023) since multi-regime tests capture spike periods more effectively than a single fixed-parameter model. To counter the classic class imbalance problem since price spikes occur 2-5% of the time but account for 90-95% of the commercial activity, future research could introduce SMOTE to artificially expand minority class (Chawla *et al.*, 2002). Probabilistic forecasting could also be deployed as traders value quantiles over point predictions (Browell and Gilbert, 2022).

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Appendix A – Horizon scan of relevant literature on electricity price forecasting

Description	Model	Variables	Train/Test period
(Browell and Gilbert, 2022)– GB imbalance market	• Kernel Density Estimation and linear regression • Proprietary probabilistic forecasting tool	Dependent: System Price Input: generator and interconnector schedules, NESO actions, bids and offers, intraday market order books, weather forecasts and recent imbalance volumes and prices.	Train: Jan 2017 – Oct 2020 Test: Oct – Dec 2020
(Lucas <i>et al.</i> , 2020) – GB imbalance market	• Gradient Boosting • Random Forest • XGBoost	Dependent: System Price Input: Five ahead-of-time Loss of Load Probability (LOLP), five corresponding de-rated margins, production, solar and wind generation, weekdays, months, system load, day-ahead price, imbalance volume	90-10% train-test split for 2019 full-year data
(Bunn, Inekwe and Macgeehan, 2021) – GB imbalance market	• Markov Regime-Switching (non-linear) • Benchmark: Linear model, and GARCH linear model (time series model)	Dependent: System Price and its sign Input: Lag of two periods for System Price, NIV, De-Rated margin, wind and solar error (difference between actual wind generation and day-ahead forecast), non-balancing mechanism short term operating reserve, change in interconnector import-export netted volumes, demand error (difference between actual national demand and day-ahead demand forecast)	Train: Jul 2016- Jun 2019
(Yılan and Beykent, 2025) – Day-Ahead Electricity Price Forecasting in Turkish Electricity market	• XGBoost conoutperformed Support Vector Machines, Long Short-Term Memory, Random Forest, Gradient Boosting, with a R^2 of 94.12, MAE of 144.8 TRY/MWh and RMSE of 201.8 TRY/MWh	Dependent: Day-ahead market price Input: Total electricity demand, generation by source, temperature, calendar indicators (weekday/weekend, holiday flags)	Train: Jan-Sep 2023 Test: Oct - Dec 2023
(Magalhães <i>et al.</i> , 2024) – Short-term power load forecasting in Portugal	• Random Forest outperformed Auto Regressive Moving Average (ARMA), ARIMA, Artificial Neural Networks (ANN)	Dependent: Day-ahead hourly load forecast Predictive: Load information from 168hours before, Day of week and hour of day, temperature	Train: Jan 2018 – Dec 2021 Test: 2022

	and XGBoost, with a MAPE of 10.96%		
(Salem <i>et al.</i> , 2019) – Intra- hour imbalance forecasting for the Norwegian system operator	• Quantile regression forests (Random Forest)	Dependent: Imbalance volumes Predictive: Historical imbalances, temporal features (month, day of week, hour, minute, holiday flag, solar elevation), market prices, production plans, temperature, wind and river flows, planned import and export of power	Train: 12 months (2015-2016) Test: 1 month

Appendix B – Summary of source and treatment of features

Variable Category	Variable	Drop / Retain	Source	Justification
Demand Forecast and Drivers	Demand-forecast	Retain	(NESO, 2025)	Selected as the primary demand signal.
	Demand-outturn	Drop		Multicollinearity: High correlation with <i>Demand-forecast</i>
	Nd (net demand)	Drop		
	Tsd (Transmission system Demand)	Drop		
	England Wales demand	Drop	(Elexon, 2025b)	Fundamental driver of demand and generation efficiency.
	Temperature	Retain		
Generation (Renewables)	Embedded wind generation	Drop	(NESO, 2025)	Multicollinearity: High correlation with wind.
	Embedded wind capacity	Drop		Low Predictive Power: Static capacity does not capture dynamic price shifts.
	Embedded solar generation	Drop		Multicollinearity: High correlation with Solar.
	Embedded solar capacity	Drop		Low Predictive Power: Static capacity does not capture dynamic price shifts.
	Wind offshore	Merged	(Elexon, 2025b)	Combined into aggregate Wind variable.
	Wind onshore	Merged		
	Solar	Retain		High correlation with derated margin
	Generation (Conventional/ Forecast proxy)	Biomass		Retain
Fossil Gas		Retain		
Hydro pump storage		Retain		
Hydro run of river and poundage		Retain		
Fossil hard coal		Drop	Removed due to UK's complete elimination of coal generation in 2024.	
Nuclear		Retain	Base load generation profile.	
Other		Retain	For all remaining generation types.	
Pump Storage Pumping		Retain	Captures demand from storage assets.	
System Balance & Margin	Derated Margin	Retain	Primary indicator of system capacity margin/stress.	

Market Mechanics	Loss of load probability	Drop		Multicollinearity: Redundant signal to Derated Margin
	Sumpn (physical notification)	Retain		Baseline intended generation schedule.
	Market index price	Retain		Primary indicator of baseline wholesale price.
	Gas price	Retain	(Elexon, 2025c)	Key input cost driver for marginal generators.
	Interconnector net generation	Retain	(Elexon, 2025b)	Captures import/export balance impact.
	System price	Target		The dependent variable (Target).
	Net imbalance volume	Drop		Data Leakage: Value is determined ex-post (after the event).
	NOAV total accepted volume per period	Drop		Data Leakage: Balancing actions occur after gate closure.
	NBAV total accepted volume per period	Drop		
	NOAV proportion of short period volumes to long	Drop		Data Leakage: Derived from balancing actions.
	NBAV proportion of short period volumes to long	Drop		

Appendix C – Linear Regression diagnostic tests

Table C1. Performance metrics for LASSO (Transformed) model

	Train	Test
R²	0.5029	0.3339
MAE	43.05	25.73
Coefficients of the 10 selected features:		
dem_forecast_mw		-1.402331
deratedmargin_mw		1.784180
gen_interconnector_mw		-4.781692
gasprice_log1p_ppmwh		0.222910
net_physnotif_mw		-0.791268
gen_gas_log1p_mw		-16.767068
gen_pumped_log1p_mw		4.334164
gen_other_log1p_mw		11.314524
gen_forecastwind_log1p_mw		-1.038551
marketindexprice_shiftlog1p_ppmwh		90.211811

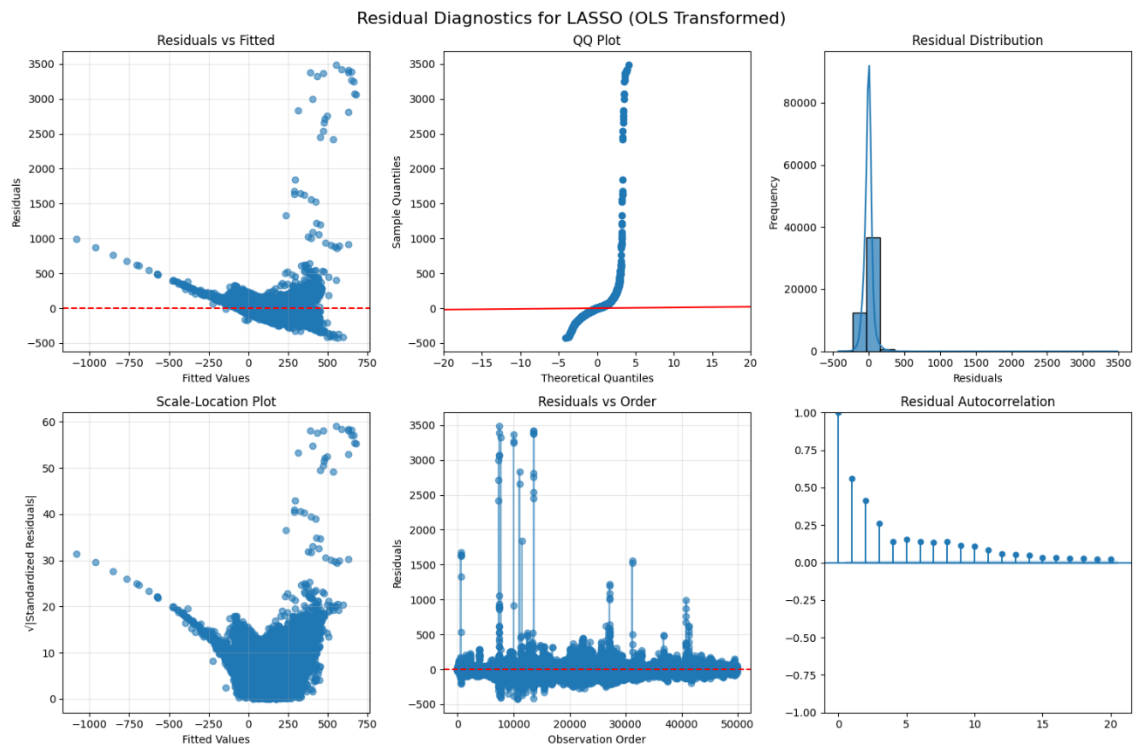
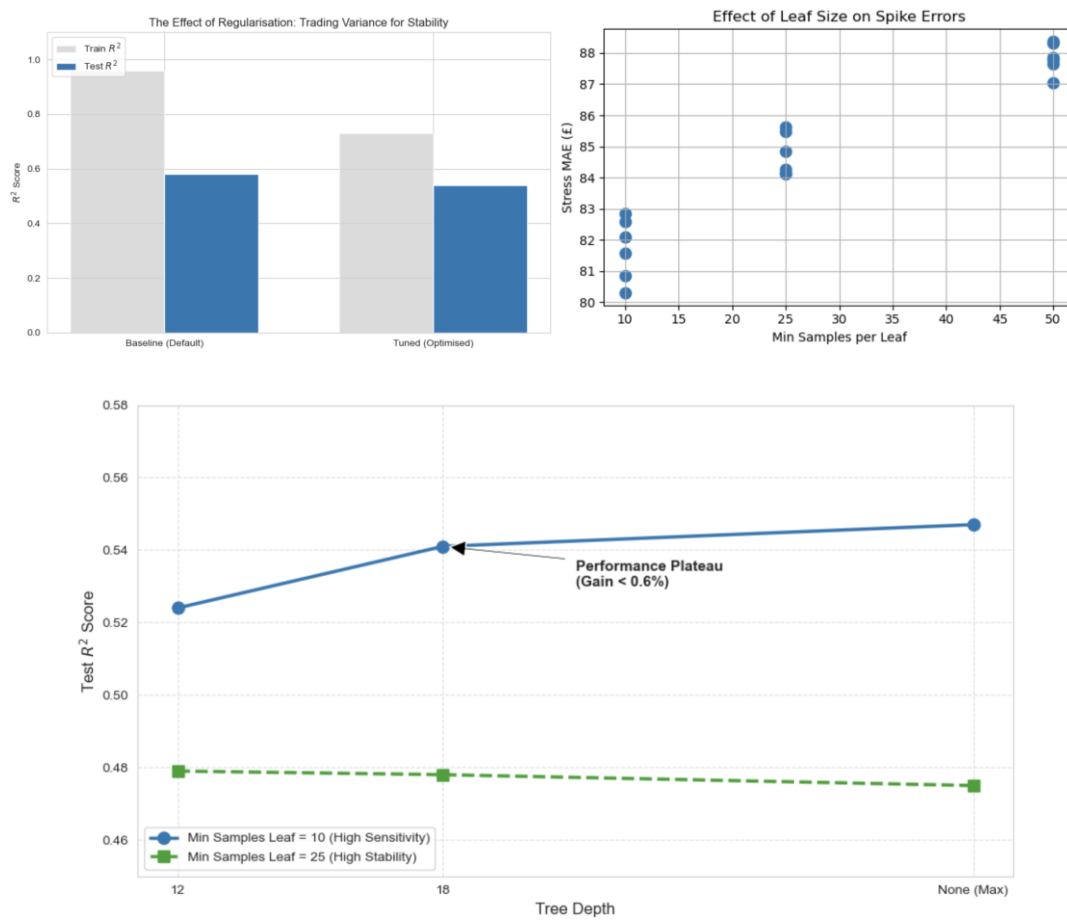


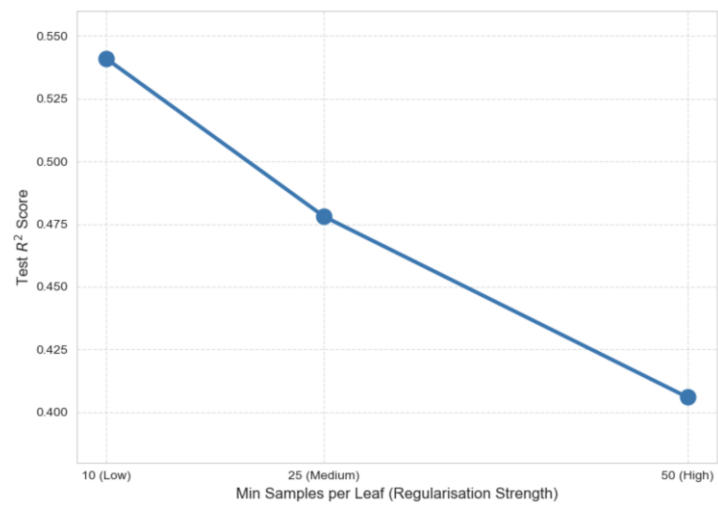
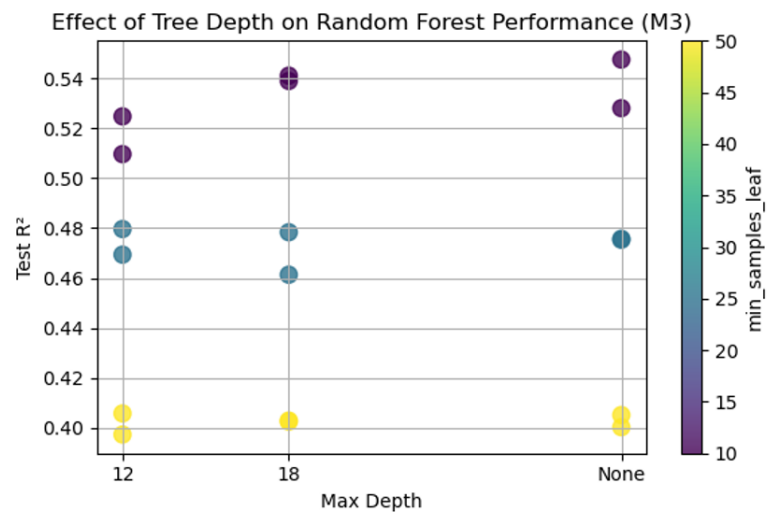
Figure C2. Residual diagnostics of LASSO (Transformed) model

Table C3. Comparison of diagnostic tests between OLS (Transformed) and LASSO (Transformed) models

Diagnostic Test	OLS	LASSO	Implications for Both Models
Durbin-Watson	0.878	0.878	Strong positive autocorrelation: Values are significantly below 2, indicating that consecutive errors are positively correlated. This is a common issue in time-series data and means the model is not fully capturing temporal dependencies.
Jarque-Bera Test (p-value)	0.000 (Reject H0)	0.000 (Reject H0)	Severe non-normality: The p-values are 0.000, clearly rejecting the null hypothesis of normally distributed residuals. Both models show extremely high skewness (OLS: 17.749, LASSO: 17.749) and kurtosis (OLS: 585.679, LASSO: 585.679), pointing to a distribution with many extreme outliers. This impacts the reliability of statistical inferences (p-values, confidence intervals).
Breusch-Pagan Test (p-value)	0.000 (Reject H0)	0.000 (Reject H0)	Heteroscedasticity: The p-values are 0.000, rejecting the null hypothesis of homoscedasticity. The variance of the residuals is not constant, meaning the errors are larger for some predictions than others. This leads to inefficient coefficient estimates and unreliable standard errors.
Rainbow Test (p-value)	0.000 (Reject H0)	Not directly applicable (N/A)	Non-linearity: For OLS, the p-value of 0.000 indicates a significant departure from linearity. This means a simple linear relationship is likely insufficient to model the underlying patterns. While not directly applicable to LASSO in the same way, the persistent issues in other diagnostics suggest non-linear relationships remain unaddressed.

Appendix D – Random Forest Grid Search Analysis





Appendix E - XGBoost Grid Search Analysis

